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UNIVERSAL DYNAMICAL RESPONSE TO SLOW DRIVING IN CHAOTIC SYSTEMS

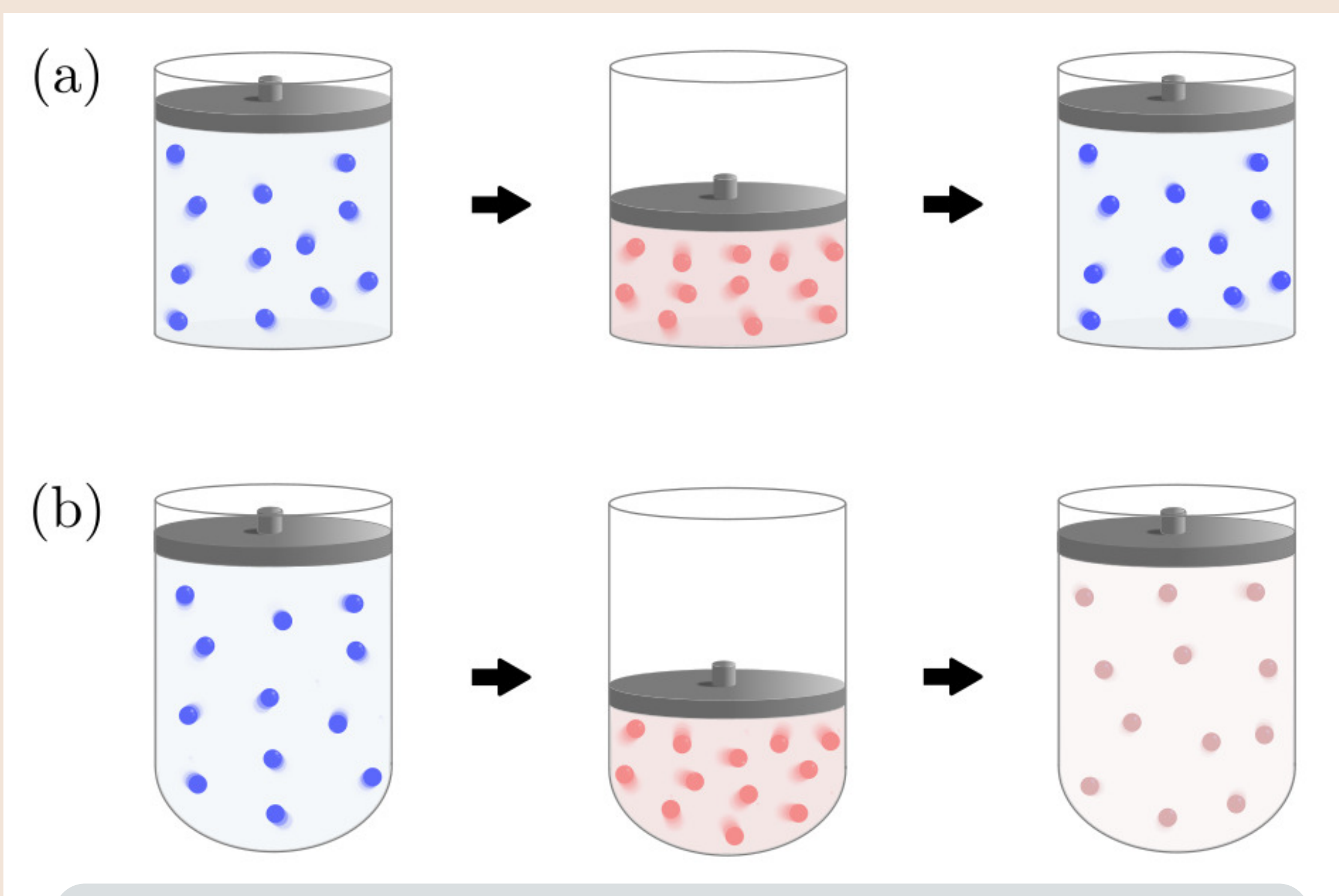
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Nachiket Karve, Nathan Rose, David Campbell, and Anatoli Polkovnikov

Department of Physics, Boston University, Boston, Massachusetts 02215, USA

Main Idea

Current notions of classical and quantum chaos are disparate. We instead focus on experimentally accessible probes. In particular, **both classical and quantum chaotic systems are sensitive to slow driving, under which they may exhibit irreversible heating.**



Regular vs chaotic piston geometries.

Framework

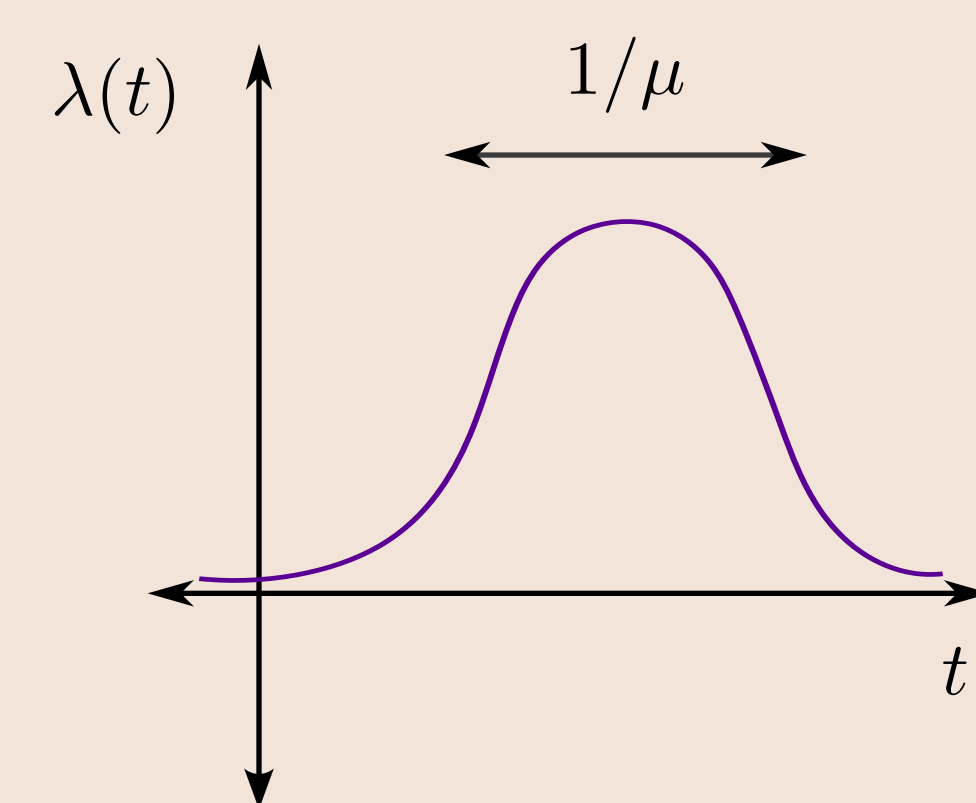
Consider a cyclically perturbed system with the Hamiltonian

$$H(t) = H_0 + \lambda(t)V, \quad \lambda(\pm\infty) = 0.$$

Define the average speed of the perturbation as

$$\bar{v} = \mu \int_{-\infty}^{\infty} dt \left| \dot{\lambda}(t) \right|,$$

where $1/\mu$ is the duration of the perturbation. The quasi-static limit corresponds to $\mu \rightarrow 0$.



Perturbation time-profile.

We compute the fidelity $\mathcal{F}$ between the initial and final states $\rho_{\pm\infty}$ and define the **speed-Fisher information** as the fidelity susceptibility with respect to the average speed:

$$\mathcal{I}_{\bar{v}}(\mu) = -2 \frac{\partial^2 \mathcal{F}}{\partial \bar{v}^2} \Big|_{\bar{v}=0}.$$

Claim: Chaos $\implies$ large $\mathcal{I}_{\bar{v}}(\mu \rightarrow 0)$

In a Gibbs ensemble, the speed-Fisher information depends on the spectral function of the perturbation V , defined as the Fourier transform of the symmetric, connected auto-correlation function:

$$\mathcal{D}_V(t) = \frac{1}{2} \langle [V(t), V(0)]_+ \rangle_c.$$

Low-frequency spectral weight controls the slow-driving response, and allows us to classify systems into three categories:

	$\tilde{\mathcal{D}}_V(\omega \rightarrow 0)$	$\mathcal{I}_{\bar{v}}(\mu \rightarrow 0)$
Regular	$\sim \omega ^\alpha \ (\alpha > 1)$	$\mathcal{O}(\mu^0)$
Chaotic thermalizing	$\text{const} > 0$	$\mathcal{O}(1/\mu)$
Chaotic nonthermalizing	$\sim \omega ^\alpha \ (\alpha < 0)$	$\mathcal{O}(1/\mu^{1+ \alpha })$

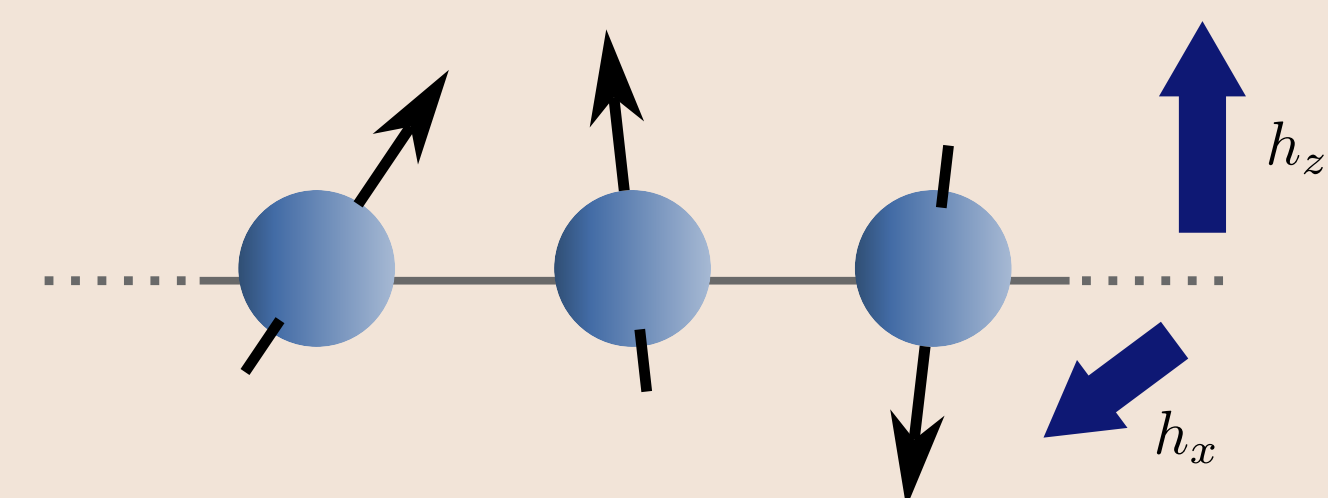
The Fisher information can be measured via irreversible heating in a chaotic system:

$$\beta \langle \Delta E \rangle \approx \frac{1}{2} \mathcal{I}_{\bar{v}}(\mu) \bar{v}^2.$$

1D Mixed-Field Ising Model

$$H_0 = J \sum_{i=1}^N \sigma_i^z \sigma_{i+1}^z + h_x \sum_{i=1}^N \sigma_i^x + h_z \sum_{i=1}^N \sigma_i^z.$$

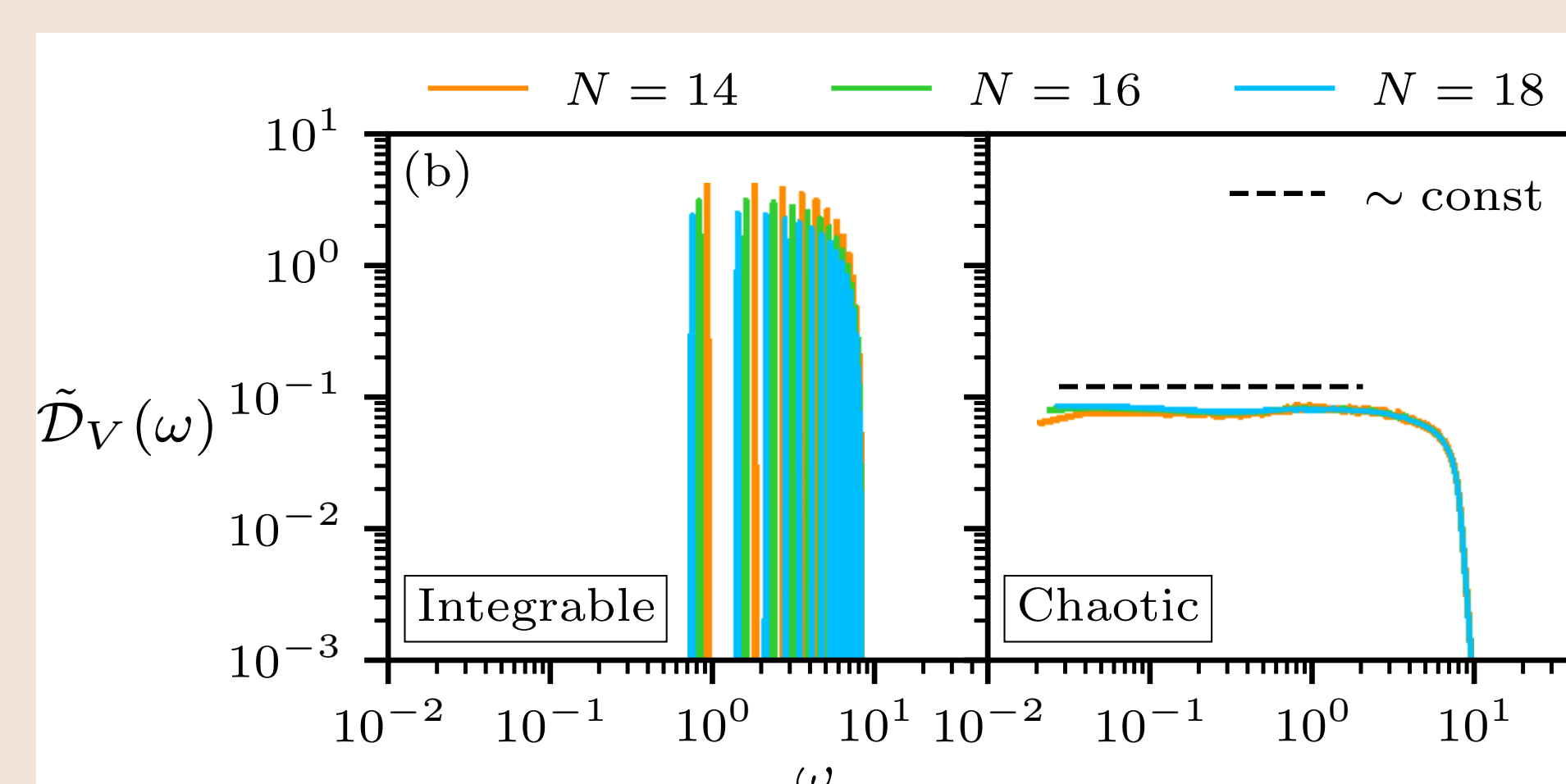
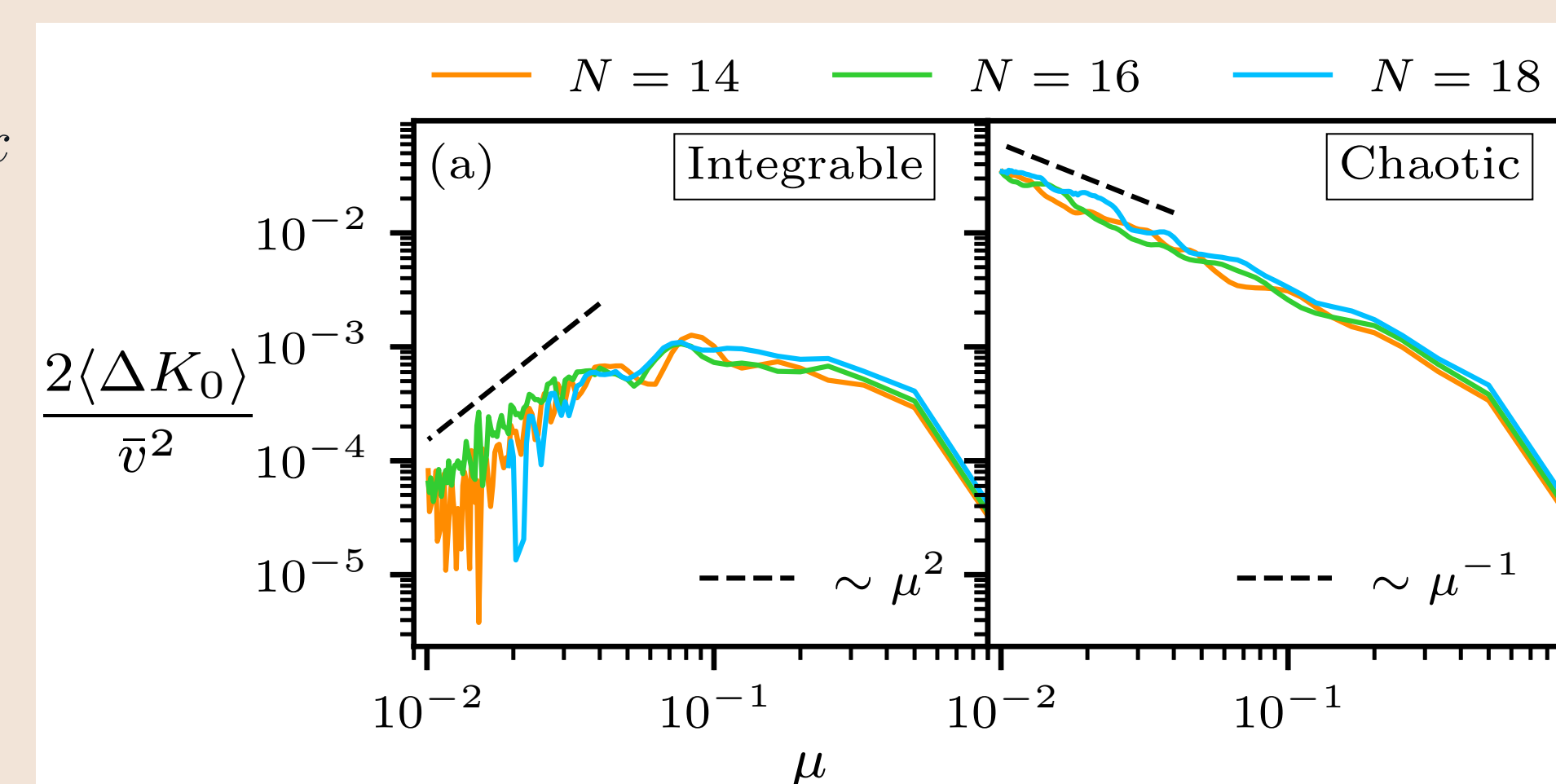
TFIM: Integrable.
Mixed-field: Chaotic.



Cyclically perturb the system through the transverse field:

$$V = \sum_{i=1}^N \sigma_i^x.$$

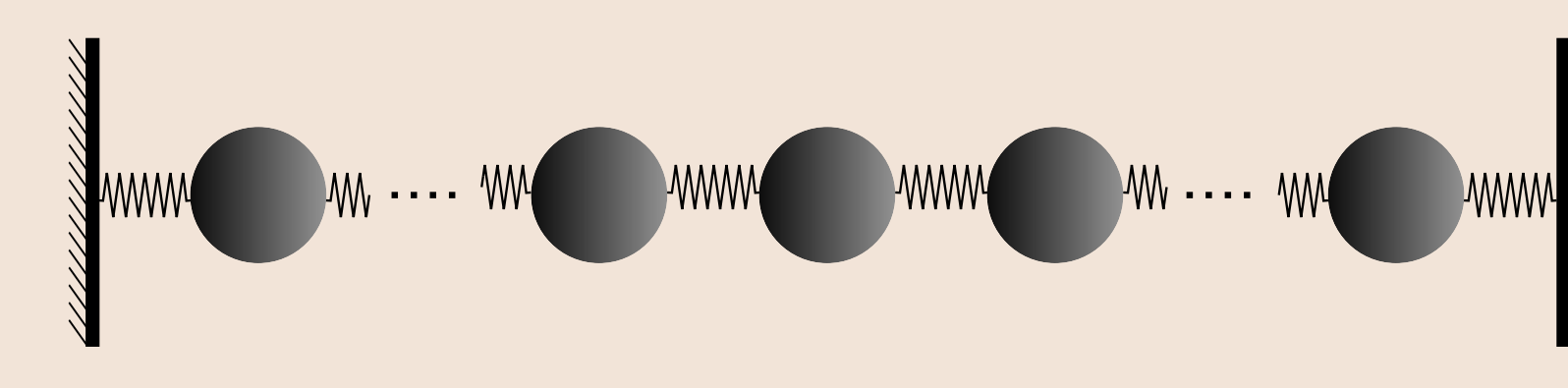
Integrable: vanishing $\mathcal{I}_{\bar{v}}(\mu \rightarrow 0)$. **Chaotic:** diverging $\mathcal{I}_{\bar{v}}(\mu \rightarrow 0)$



beta-FPUT Model

$$H_0 = \frac{1}{2} \sum_{n=1}^N p_n^2 + \frac{1}{2} \sum_{n=0}^N (q_{n+1} - q_n)^2 + \frac{\beta}{4} \sum_{n=0}^N (q_{n+1} - q_n)^4.$$

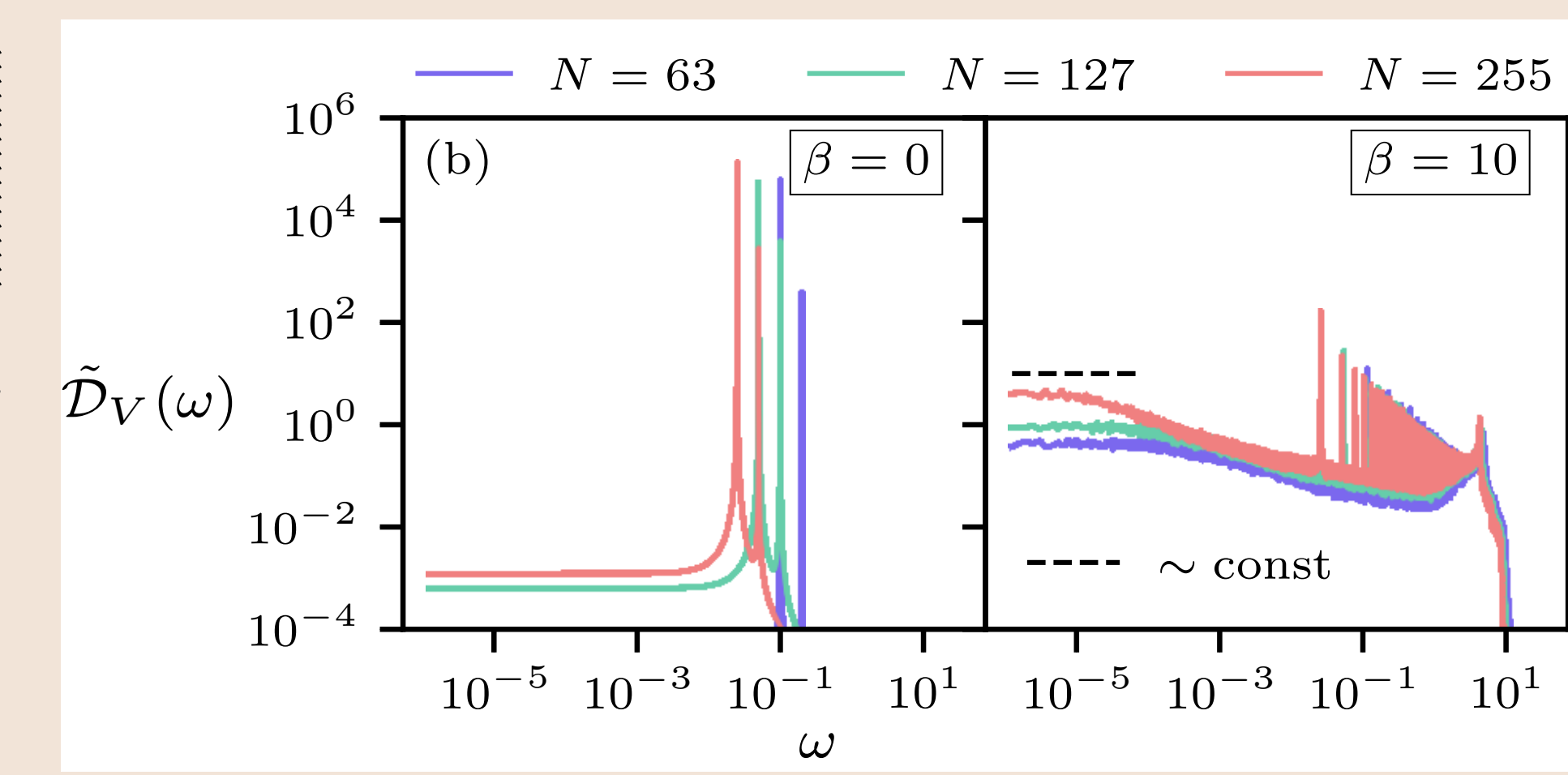
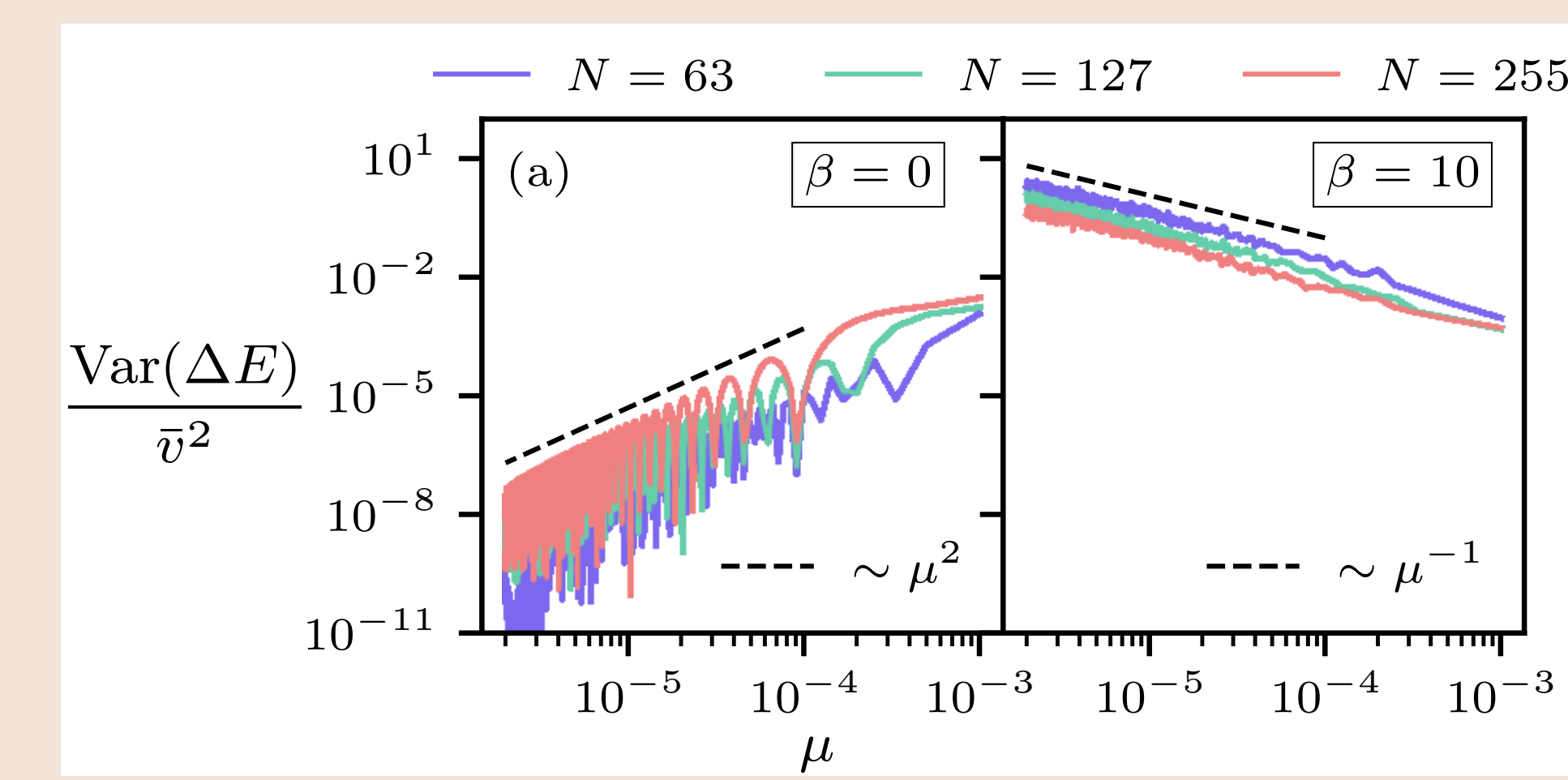
$\beta = 0$: Integrable.
 $\beta \neq 0$: Chaotic.



Cyclically perturb the system through

$$V = \frac{1}{4} \sum_{n=0}^N (q_{n+1} - q_n)^4.$$

Integrable: vanishing $\mathcal{I}_{\bar{v}}(\mu \rightarrow 0)$. **Chaotic:** diverging $\mathcal{I}_{\bar{v}}(\mu \rightarrow 0)$



Conclusion:

Slow cyclic driving provides a unified probe of chaos. Regular systems remain adiabatically stable, while chaotic systems generate irreversible drag controlled by low-frequency spectral weight:

$$\begin{aligned} \tilde{\mathcal{D}}_V(\omega \rightarrow 0) &> 0 \\ \Downarrow \\ \mathcal{I}_{\bar{v}}(\mu \rightarrow 0) &\rightarrow \infty \end{aligned}$$